MIL-R-11038E

14 June 1973
SUPERSEDING
MIL-R-11038D
11 January 1965

MILITARY SPECIFICATION

RANGES, OIL-BURNING (HEAVY DUTY, MECHANICAL TYPE BURNERS, NO. 2 OR HEAVIER FUEL OIL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers griddle top, heavy duty oil-burning ranges with a rotary-cup, mechanical-atomization-type burner.

1.2 Classification. The ranges shall be the following styles, as specified (see 6.2):

Style 1 - Right hand fire box.
Style 2 - Left hand fire box.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.
VV-F-815 - Fuel Oil, Burner.

MILITARY

MIL-G-858 - Gray Iron Alloy; Castings (Scale Resisting).
MIL-R-11337 - Ranges, Packaging and Packing Of.

FSC 7310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Sanitation Foundation (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment.

Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

Underwriters' Laboratories, Inc. (UL)

Gas and Oil Equipment List.
Standard No. 296 - Oil Burners.
Standard No. 343 - Pumps for Oil Burning Appliances.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; Walt Whitman Road, Melville, L.I., NY 11749; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

American National Standards Institute (ANSI)

Standard B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10019.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.2 and 6.3).

* 3.2 Standard product. The ranges delivered under this specification shall be the manufacturer's commercial product except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Codes and standards. The oil burning ranges shall comply with the applicable requirements of NSF Standard No. 4. The oil burner shall comply with the applicable requirements of UL Standard No. 296. The oil pump shall comply with the applicable requirements of UL Standard No. 343.

3.3.1 Compliance. Prior to the approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the ranges he proposes to furnish under this specification meet the applicable requirements of NSF Standard No. 4 and the oil burners used in the ranges meet the applicable requirements of UL Standard No. 296, and the pumps meet the applicable requirements of UL 343, as follows:

3.3.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished range, or
(2) A certification for the range issued by NSF under their special one-time contract evaluation/certification service, or
(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the ranges are being procured, indicating that the ranges have been tested and conform to NSF.

3.3.1.2 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government indicating that the oil burner and the pump comply with the applicable requirements of UL Standards No. 296 and 343.

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial oil burning ranges provided the completed item complies with all provisions of this specification.

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3.4.1 Carbon steel. The carbon steel shall be standard commercial grade used for this type of application.

3.4.2 Cast iron alloy. The range tops shall be cast from the class 2 alloy specified in MIL-G-858.

3.5 Design. The ranges shall be the heavy-duty, oil-burning type, complete with burners and operating controls. The style 1 range shall be right hand fired and the style 2 range shall be left hand fired. Ranges shall be provided with one oven and a top plate with griddle and hot top sections. The burner shall be mounted on either a flush or recessed plate at the front of the range. The burner shall be designed to burn grade DF-A (arctic grade), grade DF-1 (winter grade), or grade DF-2 (regular) diesel fuel oil conforming to VV-F-800 or grade No. 1 or grade No. 2 burner fuel oil conforming to VV-F-815. No part of the burner shall project more than 13 inches beyond the main front of the range.

3.5.1 Electrical characteristics. Unless otherwise specified (see 6.2), the electrical characteristics of the oil-burner, the oil-pump and all safety controls shall be 120-volt, 60 hertz, single phase. When electrical characteristics specified are such that the 120-volt safety control circuit required by UL Standard 296 cannot be obtained without the use of a transformer, conformance to UL Standard 296 will be waived, provided the equipment furnished is identical to UL listed equipment, except for the electrical characteristics of the safety control system.

3.6 Construction.

3.6.1 Dimensions. Ranges and ovens shall comply with the dimensions specified in table I.

TABLE I. Overall dimensions of ranges and ovens in inches

Height to griddle top from floor		Width		Depth		Oven				
						Height	Width		Depth	
Min.	Max.	Min.	Max.	Min.	Max.	Min.	Min.	Max.	Min.	Max.
32	34	48	60	34	38	15	22	27 1/4	22	27

3.6.2 Range body. The range body shall be constructed of minimum 0.0613-inch thick sheet steel specified in 3.4.1. The sheets shall be riveted, braced, welded, and reinforced in accordance with the manufacturer's standard practice. Speed nuts are not allowed in construction. Four 5/16-inch steel

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studs, spaced as shown on figure 1, shall be located on the front of the range to receive the burner mounting plate (see 3.6.9). The ranges shall be mounted on four removable legs with provisions for leveling. Legs shall be tubular, formed from carbon steel or corrosion-resisting steel, or cast from grey iron. Legs shall be bolted to the body or screwed into a socket or flange welded to the bottom of the range and shall not extent beyond the width and depth of the body. When tested as specified in 4.7.1 the legs and body shall be capable of supporting a load of 600 pounds without deformation. When specified (see 6.2), ranges shall be provided with a back shelf. The shelf shall be not greater than 26 inches high.

3.6.3 Body insulation. The range bottom, other than under the fire box, shall be insulated with a minimum of 1/4-inch thick asbestos millboard. The end of the range adjacent to the oven shall have a minimum of 1/2-inch asbestos millboard. The back of the range exposed to hot flue gases, except the back of the fire box and oven, shall be insulated in accordance with the manufacturer's standard commercial practice. All interior sheet metal parts which could come in direct contact with the flame shall be protected by cast iron or refractory material. Refractory material shall be provided between the combustion chamber and the oven to protect the oven side.

3.6.4 Fire box. The fire box shall be sized and shaped to provide proper combustion with the burner operating and shall be lined with refractory material. The refractory surfaces shall be sufficient to maintain proper combustion at burner fuel rates up to 1-1/2 gallons per hour (see 3.6.9). The combustion chamber shall be insulated from the back-wall and end-wall. Proper combustion shall be checked during the oven temperature test specified in 4.7.2.1.

3.6.5 Oven. Oven sides, top, and bottom shall be constructed of not less than 0.096-inch thick sheet steel specified in 3.4.1 except that a cast iron bottom may be provided in lieu of the sheet steel specified. The back of the oven shall be sheet steel not less than 0.0613-inch thick. A space shall be provided for the circulation of hot air between the sheet or plate forming the bottom of the oven and the sheet forming the bottom of the range. Like spaces shall be provided above the oven and on the side of the oven opposite the fire box. The oven shall be properly insulated from the combustion chamber to attain and maintain uniform temperatures between 250F. and 450F. When tested as specified in 4.7.2.1 the maximum variation in thermocouple readings of all the points measured within the oven chamber shall not exceed 40 degrees.

3.6.5.1 Oven door. The oven door shall be constructed of not less than 0.0613-inch thick sheet steel over a steel frame with linings not less than 0.0478-inch thick. Material shall be as specified in 3.4.1. The door shall be downward opening, spring counter-balanced, shelf-type, flush door which when open shall rest horizontally level with the oven deck. The door shall

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be mounted on renewable bearings or hinges and shall be provided with a heavy-duty, air-cooled-type handle. The door openings shall be the full width of the oven interior with a height not less than shown for oven height in table I. The door shall be insulated against heat loss with a minimum of 1/4-inch thick asbestos millboard. When tested as specified in 4.7.4 the door when opened shall be capable of supporting a load of 150 pounds per square foot without deformation.

3.6.5.2 Oven rack. Each oven shall be provided with a removable steel rack resting on channel supports attached to the inside of the oven. Four 1-inch channel supports shall be provided, two on each side of the oven, positioned at one-third and one-half of the total oven height. Stops shall be provided to prevent the rack from being accidentally pulled out more than two-thirds of the way. The rack shall be provided with a flange or rod at the rear to prevent pans from sliding off the rack when the rack is pulled forward. When tested as specified in 4.7.5 the rack shall be capable of supporting a load of 50 pounds without the rack being dislodged or distorted.

3.6.5.3 Flues. Flues shall be provided to insure uniform circulation of heated air around the top, bottom, and sides of the oven and under the range top. The flue shall be equipped with a draft damper arranged to control circulation of the heated air. A clean-out door shall be provided and located to permit thorough cleaning of all flues.

3.6.6 Tools and accessories. Each range shall be provided with a cover plate lifter, a flue scraper, a 24 inch length of pipe to fit flue, and an adjustable draft regulator sized for insertion in the pipe. The flue scraper shall be designed to enter both the clean out door and secondary air opening of the range for the purpose of removing soot and other accumulations.

3.6.7 Range top. Range top shall be fabricated of the cast iron alloy specified in 3.4.2 and shall have two separate sections. One section shall be over the fire box and shall contain a lid and lid-ring. The other section shall be a griddle top. The hot top section shall be not less than 16 inches wide by 22 inches deep. The griddling section shall be not less than 23-3/4 inches wide by 22 inches deep. Hot top and griddle dimensions are exclusive of the front overhang and the area adjacent to the flue opening. The range top plate shall be stiffened with ribs and supported in accordance with the manufacturer's current commercial practice. Top plate thickness of the section over the fire box shall be not less than 7/8-inch exclusive of ribs. Top plate thickness of solid sections adjacent to the lid-ring and lid section shall be not less than 3/4-inch exclusive of ribs. Top plate thickness of all other sections shall be not less than 5/8-inch exclusive of ribs. The overall thickness of all top plates with ribs shall be not less than 1-1/2 inches. When tested as specified in 4.7.2.2 the average temperature of the coolest section of the griddle top shall be between 400F. and 475F. and the average

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temperature of the hot top section shall be between 850F. and 950F. The side rail or edge casting of the range-top and back-top castings, which includes the flue collar, shall be cast iron. The front edge of the griddle top casting shall be provided with a rolled and ground edge, with a radius of not less than 1/8-inch, finished to match the top. Top shall be finished as specified in 3.9. The top side edges of the range body shall be square and flush with the sides of the range body panels to allow for side by side installation. A guard rail shall be provided on the front of the range.

3.6.8 Grease trough and container. A removable grease trough fabricated of minimum 0.0335-inch thick corrosion-resisting steel shall be provided for draining grease and drippings from the top of the range into a removable metal container attached at a front corner of the range. The container shall be fabricated of minimum 0.0335-inch thick corrosion-resisting steel and shall be provided with baffles to prevent spillage and handles to facilitate carrying. When tested as specified in 4.7.8 the container shall hold 2 gallons plus or minus 1 pint.

3.6.9 Burner. The burner shall be an integral motor driven, rotary cup, mechanical atomization type, complete with all controls and shall have a nominal capacity of 1-1/2 gallons per hour. When tested as specified in 4.7.2.1 the burner shall be capable of heating the oven to a temperature of 450F. in not more than 90 minutes at an ambient temperature of 70F. plus or minus 10F. When tested as specified in 4.7.3 the burner shall be capable of raising one gallon of water from 75F. to boiling temperature in not more than 45 minutes. The driving of the combustion air fan and the atomizing cup by belt, chain, gears, or friction shall not be allowed. The combustion air fan and atomizing cup shall be mounted directly upon the motor shaft or its extension. The fuel shall be introduced through a tube within the hollow shaft of the motor or directly into the rear of the atomizing cup. Each burner assembly shall be provided with a strainer with a removable screen. When operating with gravity supply and burning fuel oil having a viscosity of 50 Saybolt-Universal seconds, at 75- 100F. the burner components shall limit maximum firing rate to 1.50 gallons per hour while maintaining an average oven temperature of 450F. (see 4.7.6). The burner shall be provided with a mounting plate having four 11/32-inch drilled holes spaced as shown on figure 1 (see 3.6.2).

3.6.9.1 Burner controls. Each burner assembly shall be equipped with a manual resetting solenoid oil valve installed on the burner unit between the burner hinge pin and the tubing, or other means used to introduce fuel into the rotating cup. The solenoid valve shall be installed and wired so that the flow of oil to the burner will be automatically interrupted upon failure of electric service to the burner, and the oil flow shall not re-establish itself until the valve is manually reopened. The oil and air controls shall be interlocked to insure proper proportioning of air and fuel from full capacity as required by the

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range to one-third rate. The burner motor shall be provided with an inclosed safety switch incorporating thermal overload protection. The switch shall open the ungrounded side of a two-wire service to the motor and solenoid valve.

3.6.10 Pump. The pump shall be a separate wall-mounted, rotary type, either gear, modified gear, or sliding vane-type, driven by an induction motor having the same electrical characteristics as the burner with which it is operating. The pump, when tested as specified in 4.7.7 shall be capable of raising fuel through 20 feet of vertical height and delivering not less than 3 gallons per hour or more than twice the maximum fuel rating of the burner unit (see 3.6.9) which it is intended to supply, whichever is the greater. The pump shall be equipped with a float chamber and a float for controlling the operation of the pump motor. The pump motor shall not overheat when operating continuously under conditions as specified above. The float chamber shall be of such capacity and design that the pump will be off, or stopped, for at least 33 percent of the time when the wall pump is delivering 3 gallons of fuel per hour, or twice the maximum fuel rating of the burner unit which it is intended to supply, whichever is the greater (see 4.7.6). Means shall be provided to automatically stop the pump motor, if and when, the fuel supply fails or is otherwise interrupted. The float chamber shall be equipped with an overflow connection of sufficient size to return by gravity, back to the fuel tank or supply vessel, the maximum quantity of fuel the pump is capable of delivering when operated continuously. The substitution of a second float switch or other safety switch in lieu of a large overflow connection shall not be acceptable.

3.6.11 Operation of burners and pumps. When tested under the conditions specified in 4.7.6, the burners and pumps shall operate efficiently and without difficulty.

3.7 Marking for identification. Identification marking shall be on a corrosion-resistant metal plate and in accordance with MIL-STD-130. The plate shall be fastened on the front of the range above the oil burner.

3.8 Repair and maintenance. Provisions shall be made to allow for adjustment service, and replacement of burners and pumps without disturbing or disassembling of any major component or equipment. In addition, there shall be free access to inspect, service, and adjust any other component or equipment as required for proper operation of the end item.

* 3.9 Finish. The range, excluding the top, shall be finished in accordance with the manufacturer's standard commercial practice. Metallic plating, when used, shall be uniform and free from blisters, peeling and pitting. The griddle top and hot top sections shall be machine ground and polished to a uniform finish not rougher than 63 micro inches conforming to ANSI Standard B46.1.

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3.10 Workmanship. All components and assemblies of the range shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners.

3.10.1 Metal fabrication. Metal used in the fabrication of the range shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

3.10.2 Welding. The surfaces of parts to be welded shall be free of rust, paint, scale, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts and incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.10.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall be equipped with locking or restraining devices to prevent loosening and shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the holes completely and the heads and peened ends shall be in full contact with the surfaces of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. When a first article is required the range shall be examined for defects listed in table II and for dimensions specified and subjected to the tests marked "X" in column 2 of table III. The presence of any defect or any dimension not as specified or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 In-process inspection. Examination shall be made of all insulation and refractory installation as specified in 3.6.3, 3.6.4, and 3.6.5 and welding as specified in 3.10.2 to establish conformance with specified requirements. When nonconformance is noted, correction shall be made to affected items and processes.

4.4.2 End item inspection. The inspection lot shall be all oil-burning ranges of one style offered for inspection at one time.

4.4.2.1 Visual examination. The range shall be examined for the defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Standard product	Not manufacturers standard product except for changes necessary to comply with specification	X	
Finish	Not finished where required	X	
	Not as specified	X	
	Metallic plating blistered, peeled, pitted, or not uniform	X	
Construction (applicable to all components and assemblies)	Any part missing or not specified type	X	
	Functioning component that is inoperative or will not function as intended	X	
	Any component not fastened or connected as specified	X	
	Adjustable assembly that cannot be adjusted	X	
	Adjustable assembly not properly adjusted		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Workmanship (general)	Dirty, imbedded foreign matter		X
	Burrs, slivers, rough die, tool and grind marks, dents, cracks	X	
	Sharp edges or corners	X	
Castings and moldings	Sandfins, pits, blowholes, sprues	X	
Metal fabrication	Not original quality surface finish	X	
	Kink or unspecified bends	X	
	Unsquare or untrue corners		X
	Bends not uniform in size and shape		X
Welding	Not smooth		X
	Cracks, burn holes, undercuts, incomplete fusion	X	
	Scale and flux not removed		X
Fastening devices	Fastener and rivet holes not accurately punched or drilled	X	
	Burrs not removed		X
	Threaded fasteners not equipped with locking or restraining devices		X
	Threaded fasteners broken, cracked or stripped	X	
	Loose		X
	Rivet heads and peened ends not in full contact with joined surfaces	X	
Assembly	Any component not properly assembled	X	
Marking for identification	Missing, incomplete, not in accordance with MIL-STD-130		X

4.4.2.2 Dimensional examination. The ranges shall be examined for compliance with dimensions specified. Any dimension not within the

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specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

* 4.4.2.3 End item testing. When a first article is not required the first produced range shall be subjected to the tests marked "X" in column 4 of table III. Each sample range shall be tested as specified in 4.7.4 and 4.7.5. The inspection level shall be S-4. Failure of any test shall be cause for rejection of the inspection lot.

4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.6 Inspection of preparation for delivery. Inspection to determine compliance with requirements for preparation for delivery shall be performed in accordance with MIL-R-11337.

4.7 Test methods. The test schedule shall be in accordance with table III.

TABLE III. Test schedule

Test sequence	First article	Quality conformance	First produced item	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
1	X		X	Load test for body	4.7.1	3.6.2
2	X		X	Oven	4.7.2.1	3.6.4, 3.6.5, 3.6.9
3	X		X	Top surface	4.7.2.2	3.6.7
4	X		X	Boiling	4.7.3	3.6.9
5	X	X	X	Oven door	4.7.4	3.6.5.1
6	X	X	X	Oven rack	4.7.5	3.6.5.2
7	X		X	Operational (burner and pump)	4.7.6	3.6.9, 3.6.10, 3.6.11
8	X		X	Pump capacity	4.7.7	3.6.10
9	X		X	Container capacity	4.7.8	3.6.8

4.7.1 Load test for body. A uniformly distributed load of 600 pounds shall be applied across the top of the range for a 15-minute period and

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then removed. The range shall be checked for compliance with the deformation requirements of 3.6.2. Any evidence of deformation of the body or legs shall constitute failure of this test.

4.7.2 Oven and top surface temperature tests. These tests shall be conducted at an ambient temperature of 70F. plus or minus 10F. Thermocouples shall be located at the geometrical center of the oven and at the center of the back, top, bottom, and sides, 1-1/2 inches from the surfaces of the same. Three thermocouples shall be securely attached to the hot top section. One of the thermocouples shall be attached at the center and one each 2 inches in from the front and 2 inches in from the rear. Two thermocouples shall be attached to the griddling section. The two thermocouples shall be located on the griddling section area furthest from the burner and 3 inches in from the inside corners of the flue area on the diagonals. All temperatures are to be read on a recording potentiometer. Use of deflection type instruments shall not be considered satisfactory.

4.7.2.1 Oven. The burner shall be ignited. The time required for the oven temperature to reach 450F. shall be checked to determine conformance with 3.6.9. With the burner operating for a minimum of one hour and with a stabilized oven temperature of 450F., plus or minus 10F., the oven heat distribution shall be checked for compliance with the temperature requirements specified in 3.6.5. Proper combustion shall be checked to determine conformance with 3.6.4. Any nonconformance with specified requirements shall constitute failure of this test.

4.7.2.2 Top surfaces. After one hour of maintaining an oven temperature of 450F., plus or minus 10F. thermocouple readings of the hot top section and the griddling section shall be reported to determine conformance with the requirements of 3.6.7. Any noncompliance with the temperature requirements of 3.6.7 shall constitute failure of this test.

4.7.3 Boiling test. A container with one gallon of 75F. water shall be placed on the cold range top directly above the combustion chamber. The burner shall be ignited and the time required for the water to reach 212F. shall be noted to determine compliance with the requirements of 3.6.9. Any noncompliance with the boiling and time requirements of 3.6.9 shall constitute failure of this test.

4.7.4 Oven door test. The oven door, when fully open, shall have a load of 150 pounds per square foot uniformly distributed over the inner surface and checked for conformance with the deformation requirements of 3.6.5.1. Any deformation of the door shall constitute failure of this test.

4.7.5 Oven rack test. With the rack in place in the oven, a load of 50 pounds shall be evenly distributed over a 10-inch square area at the center

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and allowed to remain for 15 minutes. The rack shall not be dislodged while the load is applied and when the load is removed the rack shall be checked for conformance with the distortion requirements of 3.6.5.2. Dislodgement or distortion of the rack shall constitute failure of this test.

4.7.6 Operational test (burner and pump). With the burner (including motor) and pump in operating position and with an ambient temperature between 70F. and 100F. and with a stabilized oven temperature of 450F. the burner shall be checked for compliance with the operating time requirements of 3.6.10, the operational requirements of 3.6.11, and the maximum fuel delivery requirements of 3.6.9. Any nonconformance with specified requirements shall constitute failure of this test.

4.7.7 Pump capacity test. The pump shall be operated in an ambient temperature between 70F. and 100F. for testing capability of raising fuel 20 feet vertically in a line to determine compliance with 3.6.10. Any nonconformance with specified requirements shall constitute failure of this test.

4.7.8 Container capacity test. The grease container shall be filled to capacity and measured to determine conformance with capacity requirements of 3.6.8. Any nonconformance with specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Preservation packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-R-11337 (see 6.2).

6. NOTES

6.1 Intended use. The oil ranges covered by this specification are intended for use in the cooking of food in Military kitchens.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Style of range required (see 1.2).
(c) When a first article is required (see 3.1).
(d) Electrical characteristics, if other than specified (see 3.5.1).
(e) When a back shelf is required (see 3.6.2).
(f) Selection of applicable levels of preservation and packaging, and packing (see 5.1).

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* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Revision changes. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 7310-0418

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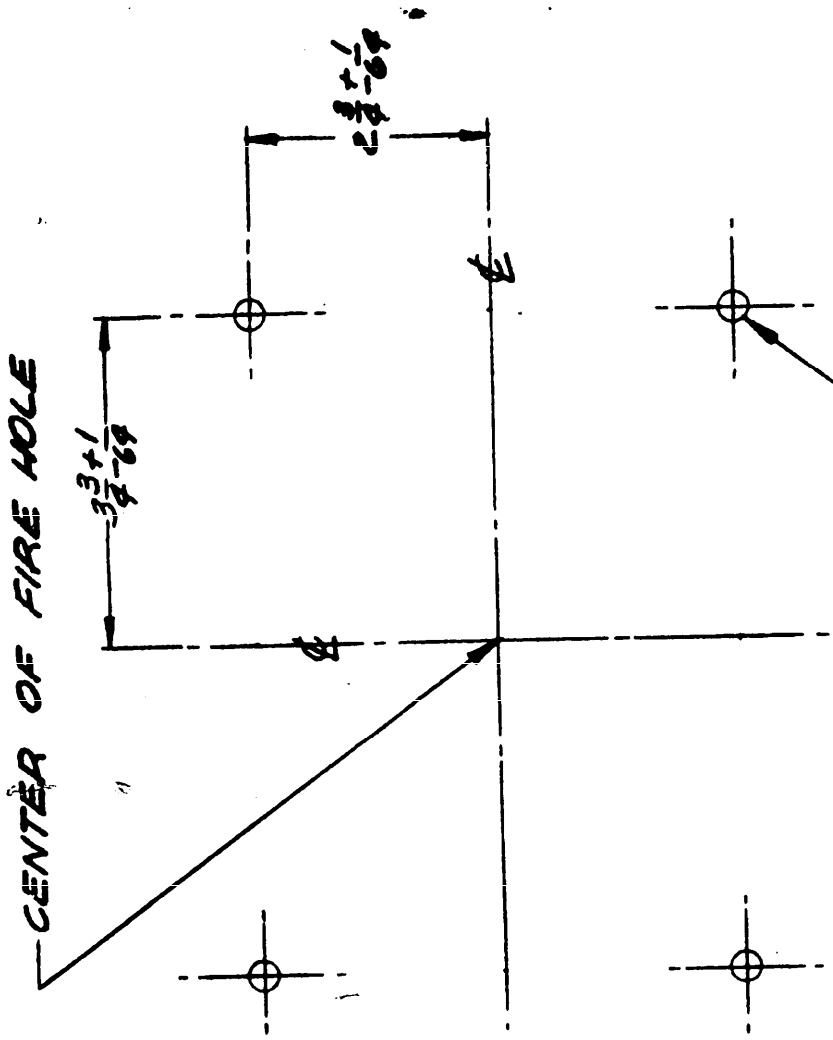


FIGURE 1. STUD LOCATION FOR MOUNTING BURNER PLATE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-11038E RANGES, OIL-BURNING (HEAVY DUTY, MECHANICAL TYPE BURNERS, NO.2 OR HEAVIER FUEL OIL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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